

ELECTROCHEMICAL METHOD OF ANALYSIS

"The methods of quantitative analysis based on the measurement of electrical properties are briefly called electrical methods of analysis."

The main techniques employed in these electrical methods are

(Mansab Gupnonnala)

A. Conductometry

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B. Potentiometry

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C. Voltametry (Polarography, Amperometry)

CONDUCTOMETRY

"It is that branch of Electro-chemistry which is based on the measurement of conductance of the electrolyte."

From a view point of electrical conduction substances are divided into two main categories

(i) Conductors :-

Those substances which permit the passage of current through them (conduct electric current), are called conductors.

(ii) Insulators :-

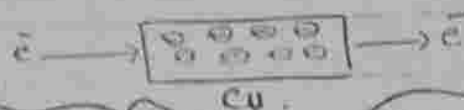
Those substances which do not conduct electricity are known as insulators.

Depending upon the nature of the charged particles present in the conductors, they can be divided into the following three

Types -

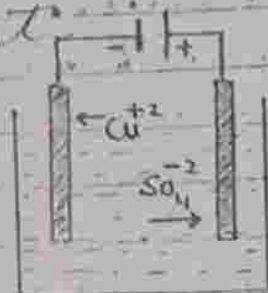
(a) Metallic or Electronic conductors :-

These are metals such as a piece of Cu or Ag. In these conductors electrons carry electric current.



(b) Electrolytic or Ionic Conductors :-

These are ionic solutions such as an aqueous soln of NaCl or CuSO_4 ; here ions are act as carriers of current.



(c) Mixed conductors :-

Those conductors in which current is partly carried by ions and partly by electrons as in the case of a soln of an alkali metal in liquid ammonia.

Basic Principle :-

According to "Ohm's Law", the current (I) flowing through a conductor is proportional to the applied Voltage (E)

$$I \propto E$$

$$I = \frac{E}{R} \quad IR = E$$

(3)

$$I = \frac{E}{R} \quad \text{--- (i)}$$

$$R = \frac{E}{I}$$

where R is constant for a given conductor and is called resistance.

Conductance =

The reciprocal of the resistance is called conductance.

$$\text{Conductance} = \frac{1}{\text{Resistance}}$$

$$L = \frac{1}{R} \quad \text{--- (ii)}$$

L = conductance of solution.

R = Resistance in ohm.

Unit: $L = \text{ohm}^{-1} = \text{mho's}$

We measure resistance of solution because there is no direct method of measurement of conductance.

Specific Resistance =

The resistance of a conductor is directly proportional to its length (l) and is inversely proportional to its area of cross-section (a)

$$R \propto l$$

$$R \propto \frac{1}{a}$$

Combining,

$$R \propto \frac{l}{a}$$

$$\frac{1}{\rho} =$$

SI

ρ = ohm $\times$ metre

$$R = \rho \frac{l}{a} \quad \text{--- (iii)}$$

where ρ (row) is a constant called specific resistance or resistivity of the

2. Example If l has unit $a = 1 \text{ cm}$, then
 $l = 1$

Thus, specific resistance of a conductor is equal to resistance of its cube whose each side is equal to the dimension of the unit cube of that conductor

$$\text{Q } \boxed{\rho = \frac{Ra}{l}} \quad \text{--- (2) (ohm cm)}$$

Specific conductance (K_v)

The reciprocal of the specific resistance (ρ) is called specific conductance (K) of the conductor (K is a Greek letter pronounced as Kappa).

$$\text{Specific conductance} = \frac{1}{\text{specific resistance}}$$

$$K_v = \frac{1}{\rho}$$

From eq. (3)

$$K_v = \frac{l}{Ra}$$

$$\boxed{K_v = \frac{1}{R} \times \frac{l}{a}} \quad \text{--- (4) } \frac{l}{a} = \text{a constant}$$

$$\text{If } K_v = \frac{1}{\text{ohm}} \times \frac{(\text{cm})}{(\text{cm}^2)} \quad | \quad K = \text{observed cond} \times x$$

$$= \text{ohm}^{-1} \text{cm}^{-1}$$

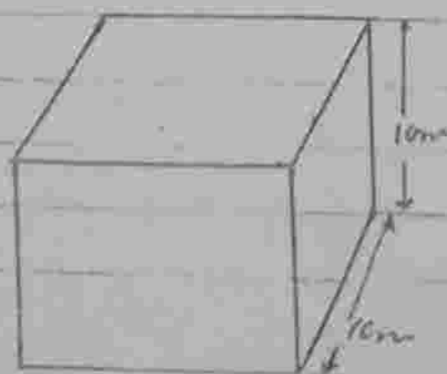


Fig: Unit cube

Specific conductance (K_s) is defined as the conductance offered by a unit cube.

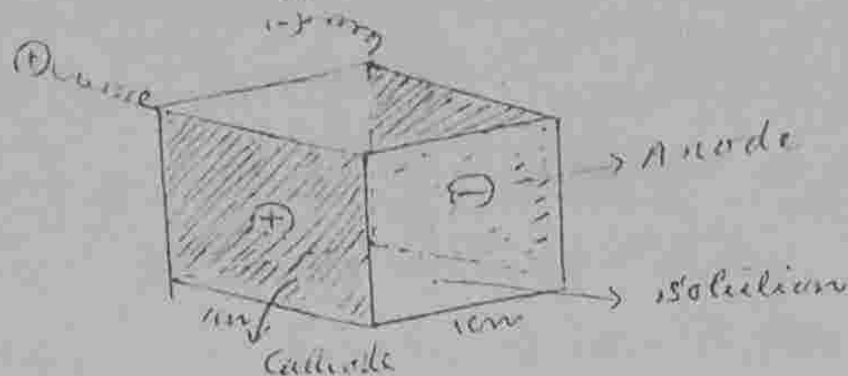


Fig. Diagrammatic illustration of of specific conductance (Measurement).

(For solid) If we have a solid conductor its unit cube (whose volume is one cubic centimetre) can be prepared and its resistance can be measured; this will be its specific resistance and its reciprocal will give the specific conductance of the conductor.

(For liquid) But if we have a conductor in the form of an electrolyte soln, a special cell, known as conductivity cell, has to be used for measuring the specific resistance of the soln conductor. A

conductivity cell is for used for performing a conductometric titration consists of two vertical platinum electrodes each of one square centimetre (1 cm^2) area and separated by a distance equal to 1 cm.

This cell is dipped into the solution whose specific conductance is to be

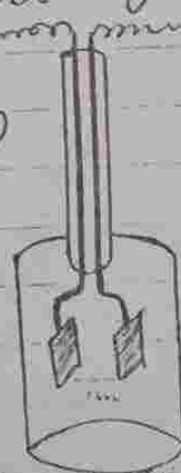


Fig. Dip type conductivity cell.

measured.

K_v = specific conductance of 1 cc solution of an electrolyte.

From the eq. (1) $\frac{L}{a}$ is called Cell constant.
Because for the measurement of specific conductance these values are remain conduct constant.

$$\text{Specific cond.} = \text{observed conductance} \times \text{cell constant}$$

$$\begin{aligned} \text{Units: } \text{specific cond} = K_v &= 1 \Omega^{-1} \text{cm}^{-1} \\ &= 1 \text{ Siemen cm}^{-1} \\ &= 1 \text{ mho cm}^{-1} \end{aligned}$$

$$\therefore \frac{1}{\Omega \text{m}} = \Omega^{-1} = \text{mho} = \text{S} = \text{Siemen.}$$

Factors affecting the specific conductance

* Specific conductance or molar conductance of an electrolyte depends upon the nature, temp and concentration.

(1) Nature of Electrolyte - (Dilution)

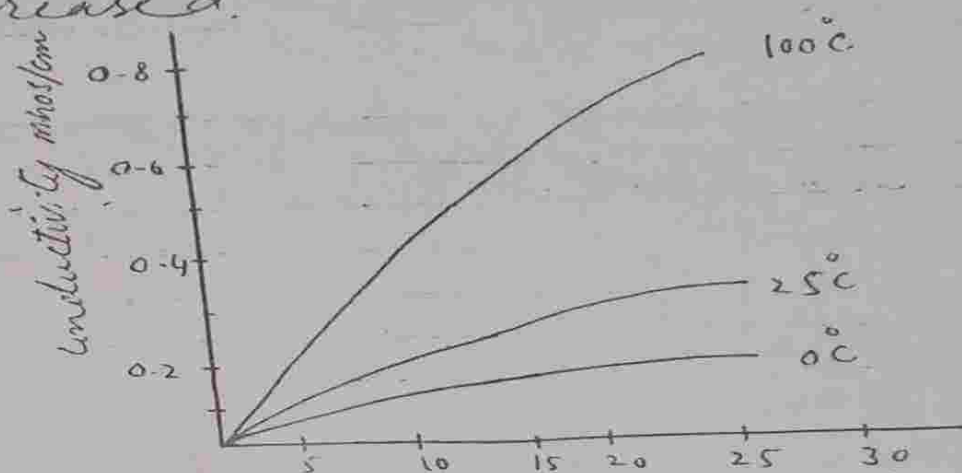
For a strong electrolytes, the conductance decrease, with dilution due to decrease in the number of current carrying ions present per unit volume of the solution.

For weak electrolytes, the value of conductance (molar eq. spx)

increases with dilution, as on dilution, more and more electrolyte ionizes.

Temperature :-

A family of the curves ~~show~~ given below, show the change of in conductivity of sodium chloride with temperature. with the increase in temp. the conductivity is also increased due to the high availability of ion carrier. And with the decrease in temp, the conductivity ~~also~~ is also decreased.



concentration present by wt.

Fig. Temp Dependence of Electrolyte's conductivity

③ Concentration

on dilution of, strong electrolyte the conductivity is decreased. and, weak electrolyte is ~~de~~ increased of weak electrolyte. The above fig. shows that specific conductance attains a maximum

values and thus decreases with the increase in concentration.

In many cases the concept of equivalent conductance and molar conductance is used instead of specific conductance.

Equivalent Conductance (λ_v)

The conductance of a solution containing 1 gm equivalent of dissolved electrolyte having two electrodes which are one cm apart.

Equivalent conductance is never determined directly, but it is calculated from specific conductance.

It is equal to the product of the specific conductance and the volume in cubic metre (m^3) containing 1 gm equivalent of the electrolyte.

$$\lambda_v = K_v \times V$$

If C is the concentration of a solution in g equivalent dm^{-3} then volume V of the solution containing 1 gm equivalent of the solute be $\frac{10^{-3}}{C} m^3$

$$\text{or } \frac{1000}{C} cm^3$$

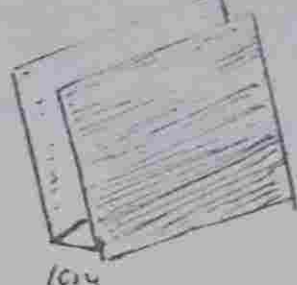
$$V = 1 \text{ gm equivalent vol} = \frac{1000 \text{ cm}^3}{C}$$

$$\boxed{\lambda_v = K_v \times \frac{1000}{C}} \quad \text{--- (5)}$$

Unit :-

$$\lambda_v = \text{ohm}^{-1} \text{cm}^{-1} \times \frac{\text{cm}^3}{\text{equivalent}}$$

$$\lambda_v = \text{ohm}^{-1} \text{cm}^2 \text{equiv}^{-1}$$



Molar Conductivity (μ_v)

The conductance of a solution containing 1 mole of an electrolyte dissolved into it certain volume V_{mol} .

$$\mu_v = K_v \times V$$

$$V = \frac{1000}{C} \text{ gm mole}$$

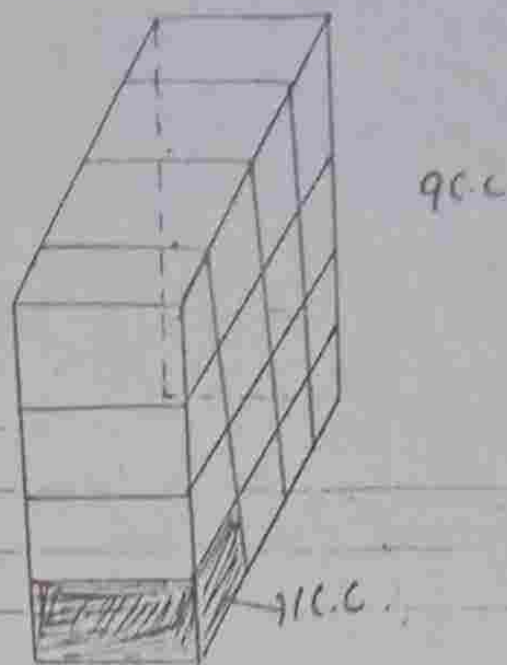
$$\text{Unit} = \text{ohm}^{-1} \text{cm}^2 \text{mole}^{-1}$$

It also depends upon Temp, nature of solute and concentration

Limiting conductance

The equivalent conductance of an infinitely dilute solution tends to be constant to a certain small value. The quantity is called the Limiting conductance.

Relationship between specific, Equivalent and molar concentration conductances



Specific conductance of a soln of an electrolyte means the conductivity of ions present in 1 c.c. of that soln.

Let us consider a rectangular prism which is divided into 9 c.c. (cubes).

Now if 1 gm. mole of electrolyte is present in 9 c.c. of its soln then conductivity due to all ions present in 9 c.c. of the soln will be equal to $9 \times K_v$. This is known

as the molar conductance of the electrolyte at given conc. and temp. and is represented by μ_v .

$$\mu_v = K_v \times V$$

Similarly: $\lambda_v = K_v \times V$ (one gm equivalent)

The equivalent conductance of an electrolyte soln, represented as λ_v , is the conductivity due to the total number of ions that can be driven from 1 gm equivalent of the electrolyte at the given dilution. The relationship between specific and equivalent conductance is given by:

$$\boxed{\lambda_v = K_v \times V} \quad \boxed{\frac{\lambda_v}{V} = \frac{K_v}{V} = K_v}$$

How Conductance is measured?

Now consider an electrolyte MA, which on electrolysis produce M^+ and A^- ions. The M^+ ions move towards cathode and gain electrons from cathode, and A^- ions move towards anode to lose their electrons at anode. So net result is a flow of electrons through electrolyte or the electricity flows through the electrolyte.

The conductance of the electrolyte depends upon the following two factors.

- (1) The total number of electrons present on cations and anions.
- (2) Velocity or mobility of ions in the solution.

There are following factors which effect on the mobility or velocity of ions.

(i) The nature of solvent (water or organic)
(polar or non polar)

(ii) ~~The nature of ion~~

If polar solvent is water then speed becomes slow due to Hydrogen bonding force. If the solvent is non polar then the mobility of ions is increased as compare to polar solvent.

(ii) The nature of ion:

The velocity of ions also depends upon the nature of ions.

e.g. Fe^{+2} and Fe^{+3} have different mobilities.

(a) charge on ion.

(b) No. of ions:

A large No. of ions also produce a hinder for their mobility.

(c) Size of ion

(iii) Size of the ions:-

(bigger) Smaller ion will rapidly move but larger ion will have slow speed.

(iv) The applied voltage.

The mobility of ions depends upon the voltage applied. If the voltage is less then ions will move slowly.

viscosity and temperature.

If the solution is viscous the mobility of ions will be less. Similarly, at low temperature mobility of ions again will be less.

(i) Concentration of solution of electrolyte.

Effect of Dilution on Specific and Equivalent Conductance

On diluting a soln of an electrolyte the number of ions per c.c becomes less and specific conductance of the soln also decreases. But as the soln is diluted, the equivalent conductance goes on increasing and eventually reaches a maximum value called equivalent conductance at infinite dilution represented as λ_{∞} or λ_0 .

There are two reasons for this increase. Firstly, on diluting an electrolyte soln, the degree of dissociation of electrolyte increases hence more and more ions become available for conductance. Since equivalent conductance includes conductivity of all the ions, it will increase with dilution of soln. Furthermore in a dilute soln

the attraction between positive and negative ions is small hence they move at faster rate in a dilute soln placed between two electrodes. This is also the reason for the increase in the conductance of an electrolyte soln on dilution.

According to Kohlrausch's law

$$\lambda_{\infty} = \lambda_{+}^{\infty} + \lambda_{-}^{\infty}$$

Kohlrausch's Law

Kohlrausch's law of independent of ionic mobilities provides the basis for conductometric principle. It states that

conductance of solution is equal to the sum of the conductance of individual ions present in soln.

$$\lambda = \lambda(\text{cation}) + \lambda(\text{anion})$$

Eg $\lambda(\text{H}_2\text{O}) = \lambda(\text{OH}^-) + \lambda(\text{H}^+)$

$$= 198.6 + 349.8 = 548.4 \text{ mho cm}^2 \text{ mol}^{-1}$$

Cation	Conductance λ^+	Anions λ^-	Conductance
H^+	349.82	OH^-	198.0
K^+	74	Br^-	78.4
Na^+	50.11	Cl^-	76.4
NH_4^+	73.4	NO_3^-	71.44
$\frac{1}{2} \text{Ca}^{+2}$	60	HCO_3^-	45
Li^+	39	$\frac{1}{2} \text{SO}_4^{+2}$	80
$\frac{1}{2} \text{Ba}^{+2}$ Ag^+	63.64 62		

Table of Ionic conductance at infinite dilution of some ions in $\text{ohm}^{-1}\text{cm}^2$
Strong Electrolytes

The conductivity of the dilute solutions of strong electrolytes increases as compared to the conc. soln. of strong electrolyte because in dilute solution there is free movement of ions and have higher velocities due to less interionic interaction whereas in highly conc. soln. of strong electrolytes, the conductance decreases due to the presence of great interionic interactions which hinder the free movement of ions through solution. e.g. 0.01N HCl has greater conductance than 11.5N HCl.

Also in H_2O , when ionizes
Conductance due to H^+ is 350

Conductance due to OH^- is 198
Because due to smaller size and H^+ ions have higher mobility than OH^- .

Similarly Na^+ have 50 and Cl^- ions have 76.

THE MEASUREMENT OF CONDUCTANCE

CONDUCTOMETRIC TITRATION

INSTRUMENTATION:

For the measurement of conductance the following sources are required.

- (1). Power source
- (2). Conductivity cell.
- (3). Resistance Bridge.

(1) Power Source

Alternating current of low intensity is used for all conductance measurements. It may have 50-10,000 Hz however most commonly is 1000 Hz.

The A.C. source may be of a low voltage tapping on the 50 Hz transformer or transistor oscillator.

A direct source current can not be used because it set up an E.M.F. of polarization which tends to oppose the flow of current.

Hall and Adams suggested a vacuum tube oscillator for conductivity measurement, in 1919. The essential circuit for any oscillator is shown.

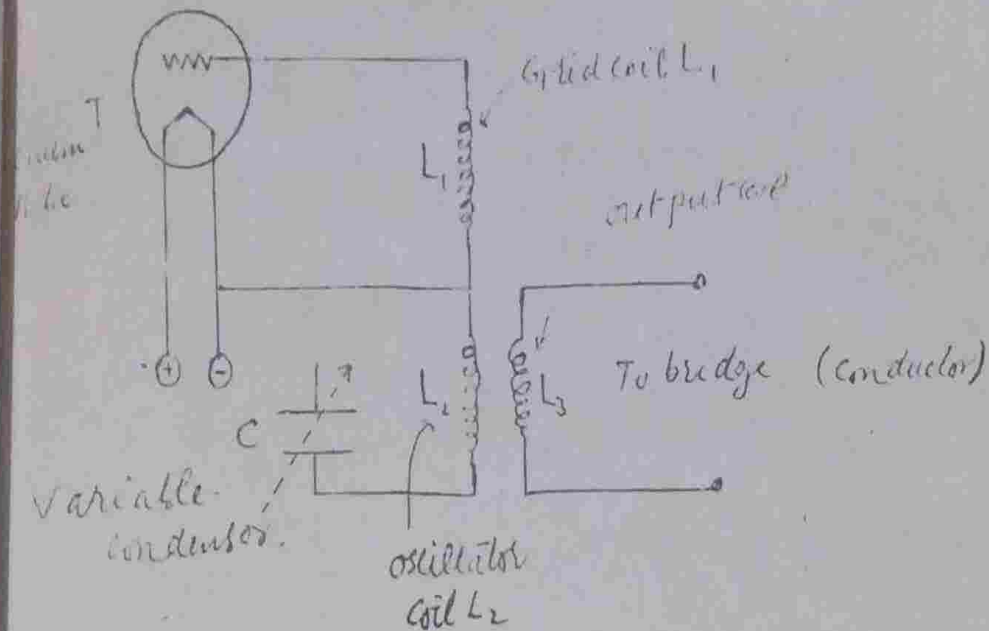


Fig. "Vacuum Tube oscillator"

The grid circuit of vacuum tube T contains a grid coil L_1 and connected to the oscillator coil L_2 in parallel with the variable capacitor 'C'. The coil L_3 which is an out put coil is coupled with L_2 inductively and conveys the oscillations to the conductance bridge.

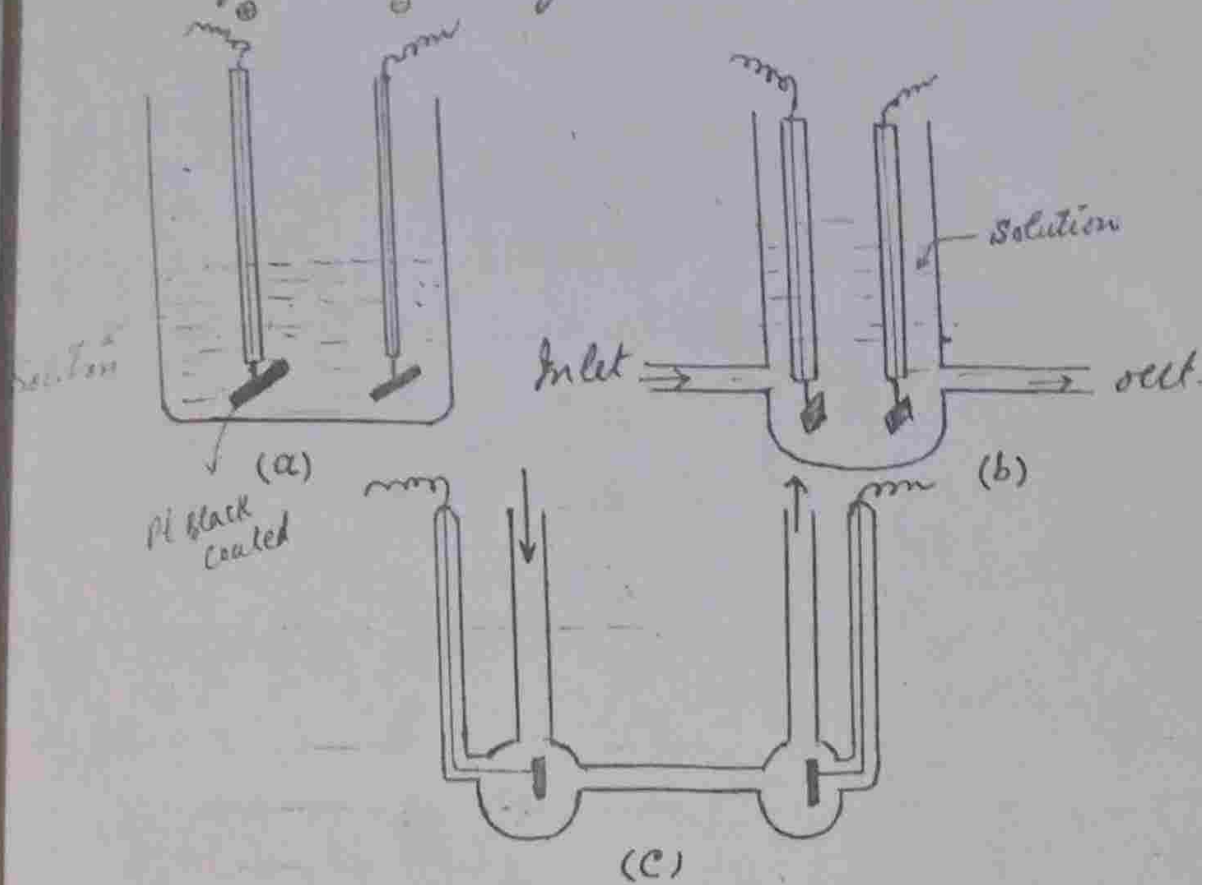
The "Vacuum tube oscillator" is better than other kinds of oscillators because it is silent in operation and produce Alternating current of constant frequency.

(2) Conductivity Cell :-

The solution whose conductance is to be measured is placed in special type of cell known as conductivity cell.

Fig(a) shows the type of cell employed for solution of low conductance

rotating figures and (c) are employed for highly conducting solutions.



The electrodes consist of Pt _{to reduce polariz} coated with finely platinum black and welded to platinum gas wires fused in glass tubes. The glass tubes contain mercury and are firmly fixed in the chomite cover of the cells so that the distance between the electrodes may not alter during the experiment.

Contact with the platinum is made by dipping the copper wires of the circuit in the mercury contained in the tubes.

(a) Electrodes. As in the above cells, Pt electrodes are used whose plates

- is coated with finely divided Pt black to reduce the polarization effect.
- 2) Conductivity water.

The conductivity water is used in the measurement of conductivity because it is sensitive for the presence of impurities and dissolved CO_2 . Distilled water due to two reasons is not used.

3) Resistance Bridge :-

The specific conductance of a solution can be determined by finding the resistance of the solution by means of Wheatstone Bridge arrangement.

The main difficulty in determining the resistance of the solution of an electrolyte is that electrolysis of the solution takes place simultaneously with the conduction of current and this produces two complications.

- (i) Polarization soon gets in and causes the resistance to vary.
- (ii) The concentration of the solution alters.

To avoid these complications, an alternating current is employed. Since under these conditions a galvanometer can not be used as in wheatstone bridge circuit, it is replaced by a head telephone

or magic eye.

If there is no flow of current through the circuit as indicated by the magic eye or head phone, the following relationship exists.

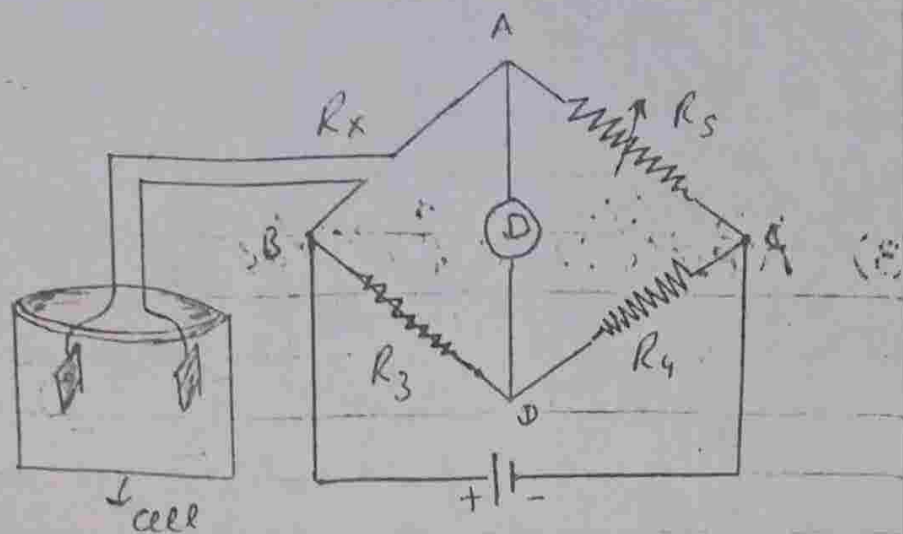


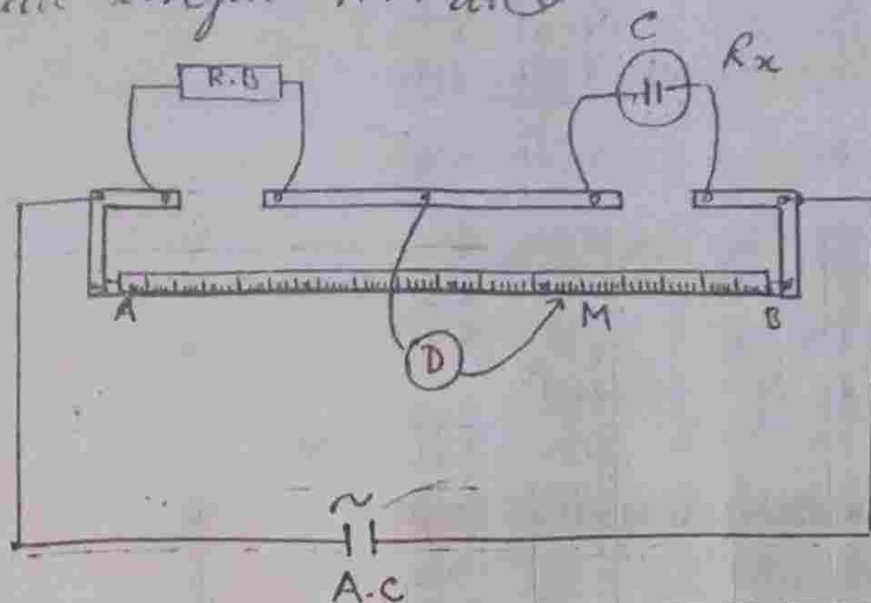
Fig: wheat stone bridge circuit.

$$\frac{R_x}{R_s} = \frac{R_3}{R_4} \quad \text{--- ①}$$

If the values of resistance of three conductors i.e. R_s , R_3 and R_4 are known, the value of the resistance for the fourth (R_x) can be calculated. The apparatus shown in the fig.

A suitable resistance is introduced by means of the resistance box and moving contact M is moved along the wire AB to a point where there is no flow of current as indicated by the head phone (at the

The length AM and BM are measured.



AB = Manganin wire of uniform thickness.

R.B. = Resistance box.

C = Conductivity cell containing electrolyte solution.

A.C. = Source of A.C. current

D = Detector magic eye or headphones

M = Moving contact (Balancing point)

Now according to Wheatstone Bridge Principle.

$$\frac{\text{Resistance in the conductivity cell } C}{\text{Resistance in the resistance box } R.B.} = \frac{BM}{AM}$$

$$\frac{R_x}{R_B} = \frac{BM}{AM} \quad \text{--- (2)}$$

Knowing the resistance of the resistance box and lengths AM and BM, the specific resistance (R_x) of the soln placed in the conductivity cell can be calculated. The reciprocal of the specific resistance will give the conductance of the solution.

$$R_x = R_B \times \frac{BM}{AM}$$

observed conductance will be as

$$\begin{aligned} \text{observed conductance} &= \frac{1}{R_x} \\ &= \frac{1}{R_B} \times \frac{AM}{BM} \quad \text{--- (2)} \end{aligned}$$

Cell constant.

As we know that

$$R \propto \frac{l}{a}$$

$$R = \rho \frac{l}{a}$$

$$R = \rho \times \kappa$$

κ is the constant factor and is called cell constant.

$$\frac{R}{\rho} = \kappa$$

$$\frac{\frac{1}{\text{observed conductance}}}{\frac{1}{\text{specific conductance}}} = \kappa$$

$$\boxed{\text{Specific conductance} = \text{observed conductance} \times \kappa}$$

Since from eq. (3)

$$\text{Specific conductance} = \left(\frac{1}{R_B} \times \frac{AM}{BM} \right) \times \text{cell constant.}$$

Determination of cell constant

Practically cell constant is measured as.

e.g. specific conductance of $\frac{N}{50}$ KCl at 25°C is 0.002765 mhos.

$$\text{cell constant} = \frac{0.002765}{\text{observed conductance}}$$

Numericals:-

- ① The specific conductance of an $N/50$ solution of KCl at 25°C is 0.002765 mhos . If the resistance of a cell containing this solution is 400 ohms , what is cell constant.

$$\text{Specific conductance} = 0.002765 \text{ mhos/cm}$$

$$\text{Resistance} = 400 \text{ ohms}$$

$$\text{Cell constant} = \frac{0.002765 \text{ (Sp. cond) cm}}{\text{observed conductance}}$$

$$\text{observed conductance} = \frac{1}{R} = \frac{1}{400} \text{ ohm}^{-1}$$

So

$$\begin{aligned} \text{cell constant} &= 0.002765 \times 400 \text{ cm}^{-1} \\ &= 1.106 \text{ cm}^{-1} \end{aligned}$$

- ② The resistance of decinormal solution of a salt occupying a volume between two Pt electrodes 1.80 cm apart and 5.4 cm^2 in area was found to be 32 ohms . Calculate the Equivalent Conductance of the solution.

Soln:-

$$\begin{aligned} l &= 1.80 \text{ cm}, \quad a = 5.4 \text{ cm}^2 \\ \text{cell constant}, \kappa &= \frac{l}{a} = \frac{1.80}{5.4} = \frac{1}{3} \text{ cm}^{-1} \end{aligned}$$

$$R = 32 \text{ ohms}$$

$$\text{observed conductance} = \frac{1}{R} = \frac{1}{32}$$

$$\text{Since the solution is } N_{10} \Rightarrow V = 10,000$$

$$\text{Specific conductance} = \frac{\text{observed conductance}}{V} \\ = \frac{1}{32} \times \frac{1}{10,000} = \frac{1}{96} \text{ mhos}$$

$$\Lambda = K \times V \quad \Rightarrow V = \frac{1000}{C} \text{ mls}$$

$$\Lambda_c = K \times \frac{1000}{N_{10}}$$

$$\Lambda_c = \frac{1}{96} \times 10,000$$

$$= 104.2 \text{ mhos} \cdot \text{cm}^2 \text{ Equi}^{-1}$$

The resistance of a conductivity cell filled with 0.01 N solution of NaCl is 210 ohms at 18°C. Calculate the equivalent conductance of the solution (cell constant = 0.88 cm⁻¹)

$$C = 0.01 \text{ N}$$

$$V = \frac{1000}{C} = \frac{1000}{0.01} = 100,000 \text{ mls}$$

$$\text{cell constant} = \kappa = 0.88 \text{ cm}^{-1}$$

$$R = 210 \text{ ohms}$$

$$\text{observed conductance} = \frac{1}{R} \\ = \frac{1}{210} \text{ ohm}^{-1}$$

$$\text{Specific conductance} = \text{observed conductance} \times \kappa$$

$$K = \frac{1}{210} \times 0.88 = \frac{1}{210} \times 0.88$$

$$= 0.0042 \text{ mhos}$$

$$\lambda = K \times V$$

$$\lambda = 0.00429 \times 100000$$

eq. conductance $\lambda = 419 \text{ mhos cm}^2 \text{ Eq}^{-1} \text{ cm}^{-1}$

Conductivity Water

It is obtained by distilling an acid solution of KMnO_4 in ordinary distilled water through an alkaline solution of KMnO_4 and condensing the vapours in a chemically resistance glass flask provided with a soda lime guard tube. The specific conductivity of such a water is near about $1 \times 10^{-6} \text{ ohms}$.

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Applications of Conductometry

There are many applications of conductance measurements. Some of these are discussed here.

(2) Solubility of Sparingly Soluble Salts

Saturated soln. of sparingly soluble salt (such that) salt is so soluble that the electrolyte present in it may be regarded as completely ionized. The relation b/w the equivalent cond. λ and specific cond. K and the concentration c is given by the eq.

$$\lambda = K_v \times \frac{1000}{c}$$

where as c represents the solubility (in gram eq. per litre).

Suppose, we have to find out the conductivity of AgCl at 25°C

The AgCl is first washed with conductivity water to remove all the soluble impurities and then AgCl is dissolved in con. water and warmed, then kept at 25°C

The solution is said to be saturated and solid AgCl will settle down

The conductivity of this solution can easily be determined. Since minute amount of AgCl is present in the solution, value of λ_v can be considered to be equal to λ_∞ i.e. Equivalent conductance at infinite dilution.

Thus we can say,

$$\lambda_v = \lambda_\infty = K_v \times V$$

where V is the volume of 1gm eq. of solute. According to Kohlrausch's law

$$\lambda_\infty = \lambda^\infty(\text{Cation}) + \lambda^\infty(\text{Anion})$$

$$= 61.92 + 76.35$$

$$= 138.27 \text{ mho's cm}^2 \text{ Eq}^{-1}$$

So from the values of K_v , λ_∞ and vol. V containing 1gm eq. can be determined and hence solubility " S " can be calculated.

$$\text{Solubility } S = \frac{1000}{V} \times E \text{ gm/e}$$

$$E = 98 \text{ g.eut.}$$

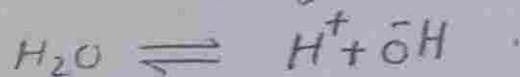
$$S = \frac{1000 \times E \times K_v}{\lambda_\infty}$$

$$S = \frac{1000 \times 143.5 \times K_v}{138.27}$$

Thus knowing K_v of the solubility of sparingly soluble salt can be calculated.

IONIC Product of water

water ionizes as



$$\therefore K = \frac{[H^+][OH^-]}{[H_2O]}$$

Since water ionizes to very small extent, concentration of water may be considered almost constant.

$$\text{i.e. } K_w = [H^+][OH^-]$$

In other words, the product of ionic concentration of H^+ and OH^- expressed in gm-moles per litre is constant at a constant temperature.

The ionic product of water can be calculated knowing the specific as well as Equivalent conductances at infinite dilution. The Eq. conductance at infinite dilution λ_∞ is represented by the Equation

$$\lambda_\infty = \frac{K_w}{c} \times 1000 \quad (K_w = 5.54 \times 10^{-8} \text{ mho/cm}^2 \text{ at } 25^\circ\text{C})$$

The value of Eq. conductance of the two ions of water at infinite dilution are

$$\lambda_\infty(H^+) = 349.8 \text{ mho cm}^2$$

$$\lambda_\infty(OH^-) = 198.6 \text{ " "}$$

$$\lambda_\infty(H_2O) = 548.4 \text{ " "}$$

Substituting the values of K_a and λ in the eq

$$K_a = \frac{K}{K_f} \times \lambda$$

$$\lambda_{\infty} = \frac{K_a}{c} \times 1000$$

we get the value of $[H^+]$ and $[OH^-]$ conc

$$\text{Thus } [H^+] = \frac{1000 \times 5.54 \times 10^{-8}}{548.4} = 1.01 \times 10^{-7} \text{ mole}$$

$$[OH^-] = \frac{1000 \times 5.54 \times 10^{-8}}{548.4} = 1.01 \times 10^{-7} \text{ mole}$$

Hence Ionic product of water

$$K_w = [H^+][OH^-]$$

$$= [1.01 \times 10^{-7}][1.01 \times 10^{-7}]$$

$$= 1.02 \times 10^{-14}$$

(3). Basicity of organic Compounds :-

After conducting the studies on the sol. salts of a large no. of organic acids, Ostwald gave empirical relation.

$$B = \frac{\lambda_{1024} - \lambda_{32}}{10.8}$$

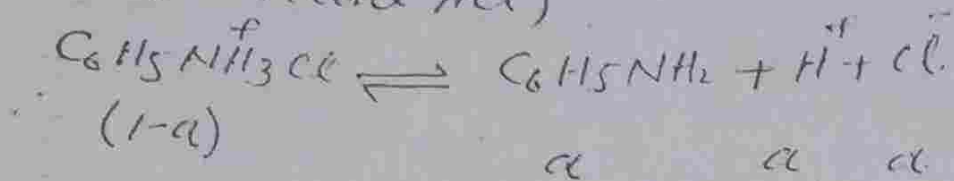
where λ_{1024} and λ_{32} are the conductance of the two salts of strength $\frac{1}{1024}$ and $\frac{1}{32}$ respectively at 25°C and B

is the basicity of the acid.

(4) Degree of Hydrolysis:-

Aniline hydrochloride which is a salt of weak base (aniline) and a strong acid

(HCl) when dissolved in water gets hydrolysed to produce $(1-\alpha)$ equivalents of unhydrolysed salt and α equivalents of free acid and free base, " α " being the degree of hydrolysis of the salt (aniline HCl)



The hydrolysis constant (K_h) can be obtained by using the relation.

$$K_h = \frac{\alpha^2 \cdot c}{1-\alpha}$$

⑤ Degree of dissociation of weak electrolyte

The degree of dissociation of a weak electrolyte is given

$$\alpha = \frac{\lambda_v}{\lambda_\infty}$$

where λ_v = eq. cond. at given dilution V

and λ_∞ = eq. cond. at infinite dilution

But the value of λ_∞ can be found from the equivalent conductance λ_{eq} at infinite dilution and λ_v can be found experimentally. Hence degree of dissociation of weak electrolyte can be calculated.

(6) Applications to Neutralization

Titration :-

Conductometric titration method can be used very successfully

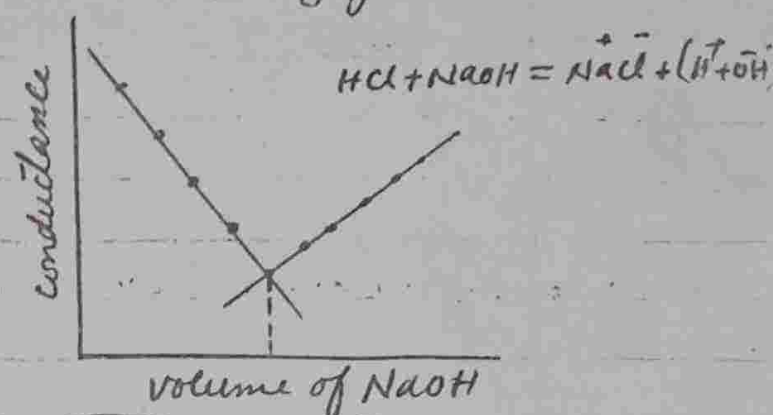
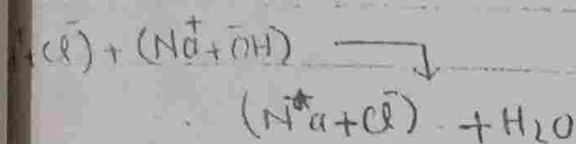
for the determination of neutralization titrations.

(i) Strong Acid Versus Strong Base (HCl + NaOH)

on adding NaOH (st. Base) to HCl (st. Acid) the more conducting H^+ ions of HCl are replaced by less conducting Na^+ ions, it means H^+ ions decreases during titration. Hence conductivity $\downarrow$ falls up to the equivalence point. After the neutralization the addition of more NaOH causes increase in the conductivity of the soln. The graph obtain for such a titration has been shown in fig.

$$\lambda(H^+) = 349.82$$

$$\lambda(Na^+) = 50.11$$

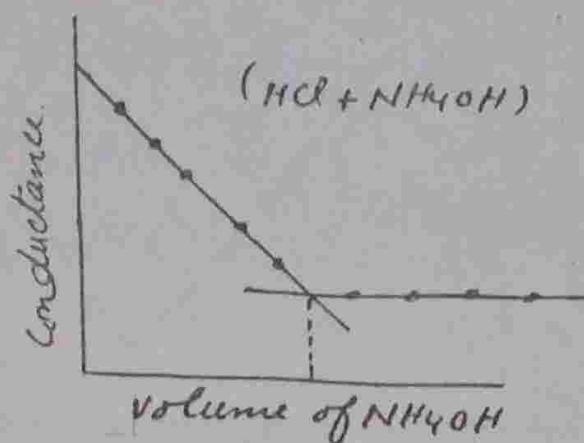


(ii) Strong Acid Versus Weak base (HCl + NH_4OH)

when NH_4OH soln is gradually added to a soln of HCl, there is replacement of highly conducting H_{350}^+ ions by less conducting NH_4^+ (73.4) ions so the conductivity of the titration liquid goes on decreasing.

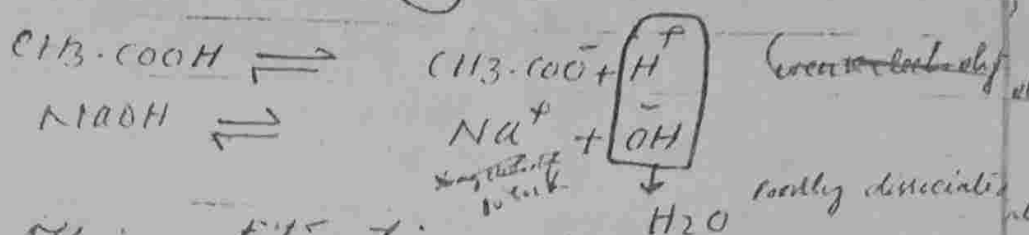
At the equivalence point all HCl is converted into NH_4Cl and further addition of NH_4OH does not cause any chemical reaction. Since NH_4OH is a weak electrolyte its addition after the equivalence point will not cause any appreciable increase in conductance (note the

the difference from casolar culture water has been used as the titrant). The graph in this case has been given in fig.



(iii) Strong acid

(iii) Weak acid versus strong base 2- $\text{CH}_3\text{-COOH} + \text{NaOH}$

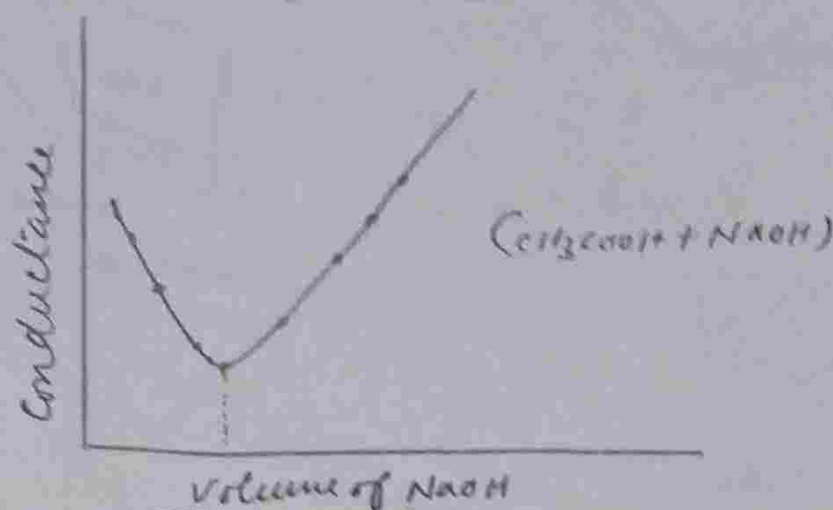


This titration gives an example of buffer whose pH remains constant and show buffer action.

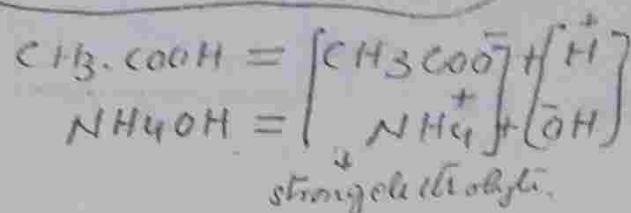
The conductance decreases over this relatively short titration zone. The establishment of the buffer solution condition stabilizes the $[\text{H}^+]$ to certain degree but with further addition of base up to the equivalence point, the conductance will gradually increase because of increase of Na^+ (50%) and [acid anion] or $[\text{CH}_3\text{-COO}^-]$.

After the conductance equivalence point a rapid increase in conductance occurs, as the $[\text{OH}^-]$ increases. The fig on the next page

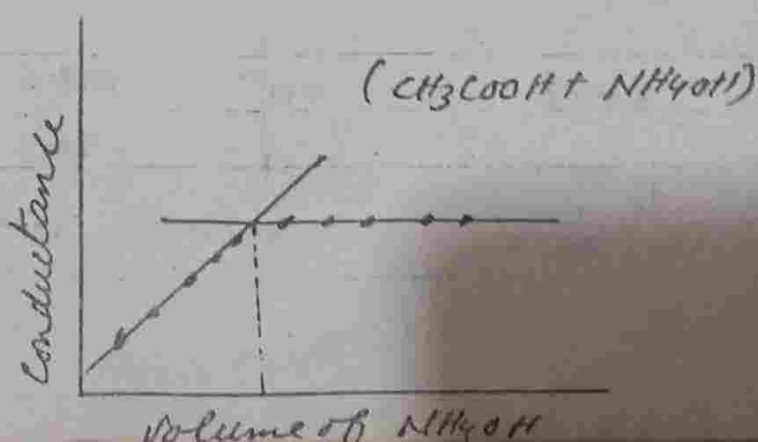
is a typical curve of this type



(iv) Weak Acid versus Weak Base ($\text{CH}_3\text{COOH} + \text{NH}_4\text{OH}$)

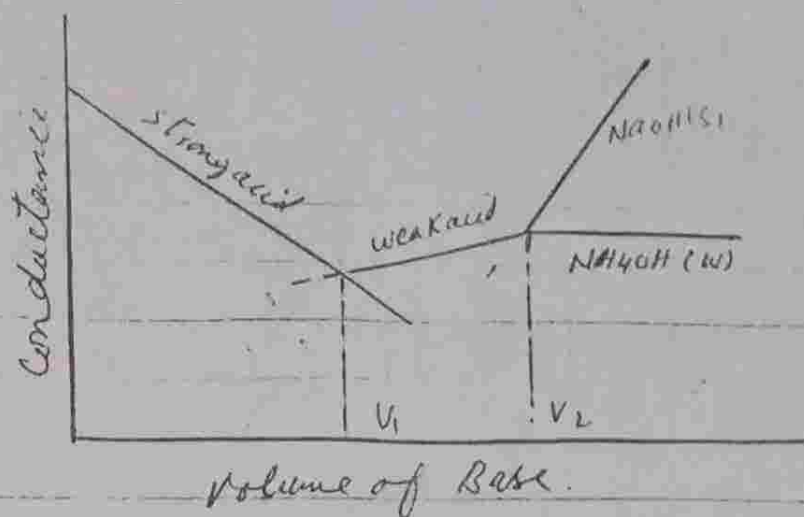


Since CH_3COOH is a weak acid, the conductivity of CH_3COOH soln. taken for titration will be quite small. On adding NH_4OH there is a formation of ~~NH_4OH~~ $\text{CH}_3\text{COONH}_4$ which is a strong electrolyte so conductance will go on increasing until the equivalence point is reached. After the equivalence point the addition of NH_4OH would not cause any appreciable change in conductance of the titration liquid because it does not undergo any chemical reaction as it itself is a weak electrolyte. Fig. shows the type of titration curve obtained in such a case.



(V) Mixture of a strong acid and weak acid ~~acid~~ Versus
strong base :-

A mixture of a strong^{acid} and weak^{acid} can be used titrated with a ~~then~~ strong base. The graphically curves will be as.

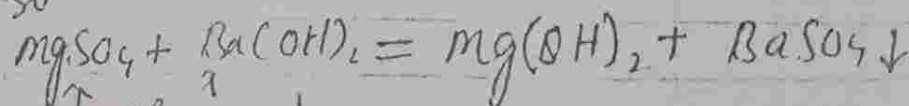


(7) Precipitation Titration :-

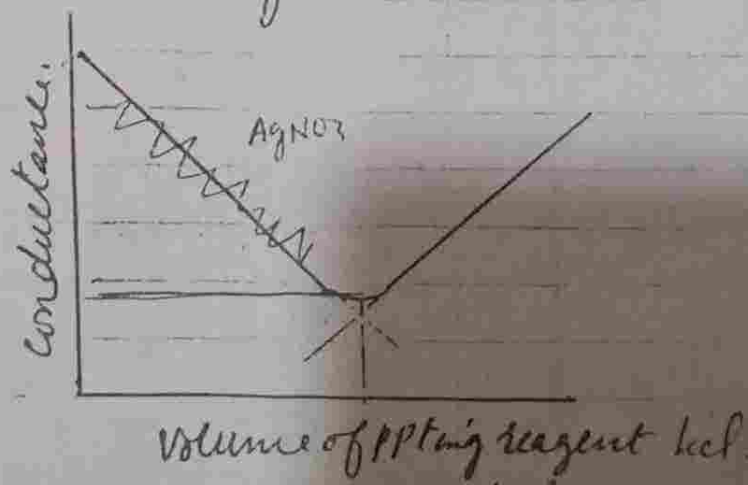
As an example of a precipitation titration, $0.1M AgNO_3$ can be titrated with $0.1M KCl$ soln.



Also

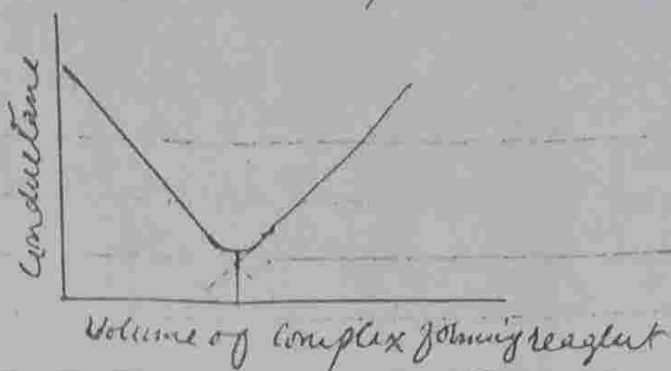


slightly soluble



Complexometric Titration

Any metal salt is
capable to form a complex with the complex
forming reagent (ligand), we
can determine the amount of
ligand which forms complex
by conductometry. The curve will
be same as in precipitation titration



(9) Determination of Vanilla in Vanilla essences

Commercial vanilla
essence contains about 1% Vanillin Vanilla,
though this conc. vary widely. By titration
the conc. of the Vanillin Vanilla in the essence
can be calculated.

At first known conc. of essence
is titrated to obtain the shape of
the curve. Then an known conc.
can be calculated.

Other Applications.

- (i) used in oceanography, measurement
of salinity of sea water.
- (ii) Estimation of moisture in wood for
pulping purposes.

Soil analysis before and after treatment.

Advand

Advantages of Cond. Titration

(i) The technique of conductometric titration can be used for the titration of coloured solutions or ⁱⁱⁱvery dilute solutions where visual indicators do not give satisfactory results.

== * ==

(iii) It gives more precisely and accurate results than ordinary classical methods. (P)

Disadvantages of Cond. Titration

(Limitations)

This technique has its limitation. These are as follows.

(i) With the increase of total electrolytic concentration it becomes accurate and less satisfactory. Actually the change in the conductance due to the addition of titrant can become largely masked by large salt concentrations in the solution being titrated, under these circumstances then this method can not be used.

- (iii) Although the method is potentially adaptable to all types of volumetric reactions, the number of useful applications to oxidation-reduction systems is limited. The reason for this is that the substantial excess of hydronium ion typically needed for such reactions tends to mask conductivity changes associated with the volumetric reaction.

Comparison between Potentiometric and Conductometric Titrations.

Similarities :-

- (i) Both are micro or semi micro techniques
- (ii) In both techniques there is no use of indicator.
- (iii) Both are not equally precise and accurate for macro levels.

Dissimilarities :-

- (i) In potentiometric methods dist. H_2O may be used but in conductometric methods always de-ionized water is used.
- (ii) In potentiometric methods solutions always are used, but in conductometry solid, solution etc. can be used i.e.

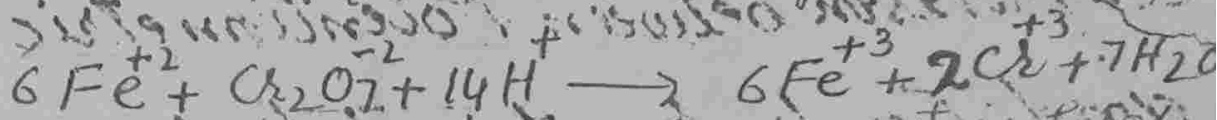
their conductance is directly measured.

Both methods are equally precise and accurate but a little difference is above.

Redox (Oxidation-Reduction) Titration:-

In case of oxidation-reduction titrations, there is a decrease in H^+ ion conc.

This is clear from the following equation



Since the mobility of H^+ is too high

therefore a sharp decrease in the

Conductance is expected during the initial part of the titration. I

has been observed that when the initial conc. of acid is moderately high, the

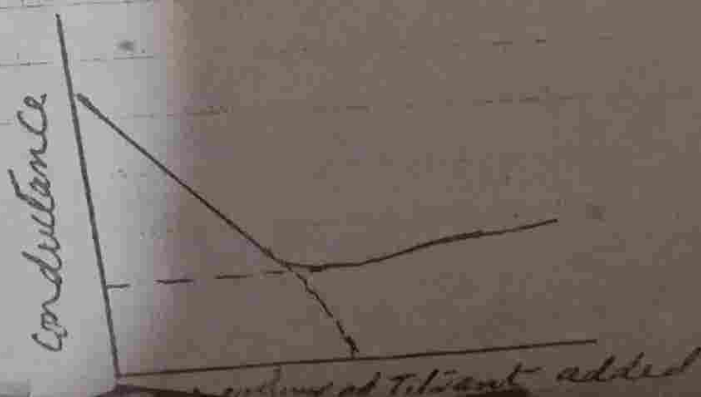
relative change in hydrogen ion conc. is

very small. The redox titrations give good results only when the initial

Conc. of acid is low. The curve for

The titration of Ferrous (iron) with dichromate in acidic medium is shown

zig.



Important Questions

No. 1 (a) Differentiate between voltammetry and polarography.

(b) Discuss the Nernst Equation and its applications in potentiometry.

(c) Describe the various types of potentiometric titrations. (P.U. 85) 8, 9, 9

Q. No. 2 (a) Discuss the principle and equipment used in potentiometry.

(b) Describe the construction of a calomel electrode and a glass electrode. (P.U. 88) 12, 13.

Q. No. 3 (a) Distinguish between Voltammetry and polarography. Define the terms "diffusion current", "Limiting current", and "half wave potential". Briefly discuss the various components of a polarograph with its circuit diagram.

(b) Discuss uses of a Dropping mercury electrode for qualitative analysis of organic and inorganic compounds. What are limitations of this electrode?

Q. No. 4 (a) (P.U. 91) 13, 12
What do you mean by.

(i) specific conductance.

(ii) Molar conductance.

(b) Describe the construction and working of a conductimetre. (P.U. 92) 10, 15

NO.5 (a) Explain the diffusion current and discuss other types of polarographic currents. what are the analytical applications of polarography.
(P.O.90) 25

Q. write notes on the followings.

(i). Conductometry and its applications (85)

(ii). Polarography. (88)

(iii) Thermal conductivity detector. (92)

(iv) Specific conductance.

(v) Various types of electrodes (90)

Q.NO.6 (a) Discuss the principle and equipment used in potentiometry.

(b) Describe the construction of a calomel electrode and a glass electrode.

(P.O.86) 12,13

Q.No.7 (a) Discuss the basic instrumentation of polarograph.

(b) what are maxima suppressors? what is their function and name any two of them.
(1087)

Q.No.8 (a) Construct and discuss the potentiometric titration curve of Iron Fe^{+3} vs any oxidizing agent.

(b) Discuss the indicator and reference electrodes used in the system. what is polarography

P.O.S. discuss the technique with special reference to (i) micro polarisable electrode (ii) use of supporting electrolyte (iii) decomposition potential (iv) capillary characteristics and (v) use of Ilkovic equation for quantitative estimation

Q NO. 9(a) Describe a simple polarographic circuit. Discuss the applications of polarography. (P.U. 89)

Q NO. 10 Describe the principles and instrumentation of potentiometry (P.U. 89)

Q NO 11

(a) Discuss Indicator and Reference electrode used in potentiometry.

(b) Discuss Briefly Three types of potentiometric titrations.

(c) What is the electrode potential?

How does Nernst Equation help in the determination of electrode potential. (P.U. 591) 13, 12

$$S \rightarrow \frac{100}{100} \quad D_1, D_2 \quad ; D = D_1 \times D_2$$

$$V_1 + V_2 = V_3 \quad \frac{100}{100} \quad 10, 10, 10 \quad \geq 1000$$

$$D = \frac{V_3}{V_1}$$

$$1 + 9 = 10$$

$$D = \frac{10}{1}$$

$$D_3 = 3 \text{ ppm}$$

$$\geq 1000 \text{ ppm} \geq 3000 \text{ ppm}$$

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$$20 \rightarrow 80 \Rightarrow 10$$

$$\frac{100}{20} = 5$$

$$I = 15 \text{ ppm}$$

$$45 \text{ ppm}$$

$$45 \text{ ppm}$$

$$u$$